

Legislative text implementation

Document generated by Catala version 1.2.1

This document was produced from a set of source files written in the Catala programming language, mixing together the legislative text and the computer code that translates it. For more information about the methodology and how to read the code, please visit <https://catala-lang.org>.

Source files weaved in this document:

- `Pilots.catala_en`, last modification 2026-09-24, 20:36

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The financial condition for a sophisticated professional investor

The source is the SFC and HKMA circular of 28 July 2023, Annex 1, paragraph 3.1. Its financial condition permits either a portfolio of at least HK\$40 million or net assets, excluding the primary residence, of at least HK\$80 million. These alternatives are only one part of the SPI requirements. The definitions of wealth, applicability and legal effectiveness still need legal review. Dates in the comparison fixtures are synthetic test settings.

The following program uses HKD cents. The first pair of inputs describes the portfolio; the second describes net assets. Each amount has a separate flag that says whether usable evidence is available. An unavailable amount cannot satisfy a threshold. One established alternative is enough; when neither is established, both must be known to return false. The result code is 1 for true, 0 for false, and -1 for unresolved. The host translates these codes into named statuses and preserves the missing evidence.

/home/chakwong/python/legalmath/.worktrees/catala/examples/catala/Pilots.catala_en

```
19 declaration scope Financial:
20   input portfolio_known content boolean
21   input portfolio_cents content integer
22   input assets_known content boolean
23   input assets_cents content integer
24   output decision content integer
25
26 scope Financial:
27   definition decision equals
28     if (portfolio_known and portfolio_cents >= 40000000000) or (assets_known and
29 assets_cents >= 80000000000)
30     then 1
31     else if portfolio_known and assets_known then 0 else -1
```

An ambiguity test for defaults and exceptions

This is a synthetic programming example, not a rule in the circular. The ordinary result is zero. Either of two exceptions changes it to ten. Catala 1.2.1 can coalesce equal exception results. RuleIR instead treats two applicable exceptions as a conflict, even when both say ten. We preserve the RuleIR interpretation by

checking ambiguity explicitly. A missing guard leaves the outcome unresolved; the amount is used only when the result code is 3. Code 2 denotes a conflict and -1 denotes missing information.

/home/chakwong/python/legalmath/.worktrees/catala/examples/catala/Pilots.catala_en

```
44 declaration scope Exceptions:
45   input first_known content boolean
46   input first_applies content boolean
47   input second_known content boolean
48   input second_applies content boolean
49   output decision content integer
50   output amount content integer
51
52 scope Exceptions:
53   label ordinary
54   definition amount equals 0
55
56   exception ordinary
57   definition amount under condition first_known and first_applies
58   consequence equals 10
59
60   exception ordinary
61   definition amount under condition second_known and second_applies
62   consequence equals 10
63
64   definition decision equals
65     if first_known and first_applies and second_known and second_applies then 2
66     else if first_known and second_known then 3 else -1
```

The amount alone demonstrates why compilation success is insufficient: two true guards produce ten natively, but the separately computed ambiguity result prevents the host from treating ten as an accepted decision.